

Do Online Communities Improve Job Performance in the Geographically Dispersed Organization?

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The past decade has witnessed the explosive implementation of corporate social tools and technologies such as corporate internal online communities. However, there is little empirical evidence on whether and to what extent the implementation of an internal online community affects individual job performance in the geographically dispersed organization. In this study, we investigate this question with a quasi-natural experiment: A large manufacturer (i.e., the company) introduced an internal online community for its sales representatives across geographically dispersed subunits. The unique data set from the company's different departments enables us to observe detailed information about each representative's community usage, sales performance, and demographics. Our difference-in-differences design and instrumental-variable analysis jointly demonstrate a positive relationship between community usage and individual sales performance. In addition, we find that the online community has heterogeneous effects: It is significantly more beneficial for representatives with fewer offline opportunities to observe and learn from their colleagues' activities (i.e., representatives from subunits with fewer employees and lower average job performance). Finally, sales performance is more positively associated with reading posts from nonlocal subunits (vs. posts from the local subunit) and posts that summarize multiple selling events (vs. posts that describe a single selling event). Our study has implications for managers regarding the benefits of corporate online communities, especially for the geographically dispersed organization.

Key words: geographically dispersed organization; individual job performance; online communities

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1. Introduction

Corporate internal online communities have become increasingly popular. According to McKinsey's 2016 report on organizational social tools, 93% of executives (compared with 82% in 2014, 73% in 2013, and 64% in 2012) report that their companies use at least one social tool or technology such as online communities for internal communication (Bughin et al. 2016, *McKinsey*; Bughin and Chui 2013, *McKinsey*). The increasing popularity of corporate internal online communities reflects a realization about the importance of knowledge sharing within the organization. Lew Platt, the former chief executive of Hewlett-Packard, eloquently summarized the importance of knowledge management: "If only HP knew what HP knows, we would be three times more productive" (Hagel and Brown 2008, *Bloomberg*). Unfortunately, knowledge moves slowly across geographic boundaries, particularly in the offline setting (e.g., Almeida and Kogut 1999, Tallman and Phene 2007), resulting in knowledge silos (Hwang and Krackhardt 2020). In the geographically dispersed organization, online communities are expected to eliminate geographic boundaries and catalyze knowledge sharing among employees from different places.

While existing literature on knowledge management systems explores the impact of online communities on employees' performances, the primary focus is the organization with one centralized location (e.g., Ko and Dennis 2011, Lu et al. 2015).¹ There is little empirical evidence on whether such systems affect individual job performance in the geographically dispersed organization. The answer is not straightforward and cannot necessarily be inferred from the existing studies. On the one hand, an online community may not necessarily help if community participants cannot communicate as well across geographic boundaries as within their own boundaries. Even though an online community eliminates physical limitations, geographically proximate (vs. distant) peers are more likely to converse (Hwang et al. 2015), and knowledge is circulated mostly among employees who are physically nearby (Hwang and Krackhardt 2020). In addition, knowledge and information that is gained in one location may not be as applicable in other locations. On the other hand, an online community provides an additional channel through which employees can observe knowledge and information that is unavailable locally (Tallman and Phene 2007). Moreover, an online community connects employees from geographically dispersed areas, heightening the

perception of being around many colleagues and possibly facilitating peer effects.

We contribute to the literature on knowledge management systems by analyzing whether and to what extent a corporate online community affects individual job performance in a geographically dispersed organization. Formally, we ask the following research question: *Does an internal online community in a geographically dispersed organization result in better individual job performance?*

We propose that the answer to this question is *yes*—an internal online community improves individual job performance in the geographically dispersed organization. A corporate online community is an additional channel through which the organization's employees can share knowledge and observe others' work without the constraint of geographic barriers. In other words, it is possible that employees can exchange their work experiences and ideas more effectively in an online community than through traditional offline communication channels. We argue that although employees are still inclined to communicate with geographically proximate peers in the online community (Hwang et al. 2015), they can better assimilate and synthesize knowledge across geographically dispersed areas more easily with an online community than without. Moreover, through an online community, employees can better sense the existence of peers and observe their peers' work experiences and performances, which can lead to a social facilitation effect or peer effect, thereby motivating employees to exert more effort (e.g., Wang et al. 2019, Zhang et al. 2017).

To empirically test the effect of a corporate online community on individual job performance, we obtain a unique data set from a large manufacturer (hereafter, "the company") that produces and distributes sauces and condiments to restaurants, hotels, and supermarkets. The company has multiple geographically dispersed marketing subunits in China. Each subunit includes sales representatives who market products and interact with customers. At the beginning of 2018, the company launched a corporate internal online community (hereafter, "the online community") for all sales representatives across the geographically dispersed subunits. The representatives can participate in the online community by writing posts about their work experiences, reading existing posts, and leaving comments on those posts. We collect each representative's detailed community usage records as well as measures of individual performance: quarterly sales for all products in 2017 and 2018 and monthly sales for one specific product (hereafter, "product-H") from August 2018 to December 2018.

We use the company's introduction of the online community as an exogenous event and design a

quasi-natural experiment to test the effect of online community on individual sales performance. We use a difference-in-differences (DID) approach combined with propensity score matching (PSM) and find a positive relationship between participation in the online community and quarterly sales. Following the existing literature (e.g., Dhanorkar and Muthulingam 2020, Jung et al. 2019, Kumar et al. 2019), we confirm the validity of the DID analysis by verifying the parallel trends assumption with a leads-lags model and a placebo test. We also perform robustness tests with several alternative matching strategies.

Moreover, we design an instrumental-variable analysis and estimate a two-stage least squares (2SLS) model to explore the relationship between reading online posts about product-H and subsequent individual sales of product-H. All posts are listed in inverse chronological order in the community, so for the instrumental variable, we use the average number of product-H-related posts that were displayed on the first post-listing webpage each time the representative logged in; this variable should be positively related to the number of product-H-related posts read by the representative but should not be related directly to the representative's sales of product-H. The results show that a representative's sales of product-H are positively associated with the number of product-H-related posts read by the representative.

We further explore the underlying mechanisms by examining heterogeneity in the effect and by comparing the impacts of different types of posts in the community. Our analyses suggest significant heterogeneity in the effect of the online community such that the community is significantly more beneficial for representatives who work at subunits with fewer employees or a lower average sales performance. We reason that the online community helps representatives improve their sales by enabling them to observe and learn from the discussions generated by others who work at geographically dispersed subunits. In addition, we show that individual job performance has a stronger positive relationship with reading posts generated by nonlocal colleagues (vs. posts generated by local colleagues) and with reading posts that summarize multiple selling events (vs. posts that describe one selling event).

To the best of our knowledge, this work is the first to examine the effects of an online community on individual job performance in a geographically dispersed organization. Using a quasi-natural experimental design, we conduct a robust empirical analysis of individual sales performance *before* and *after* the introduction of the online community. Our results indicate that individual job performance, operationalized as product sales, significantly improves

with the representative's participation in the online community. Our study contributes to the literature on (i) knowledge sharing and performance and (ii) online communities and knowledge management systems. Moreover, our results have important managerial implications. Based on our findings, companies can better assess the potential benefits of an internal online community. Our additional analyses indicate that the benefit brought by the online community is related to the number and type of posts read by the sales representatives—in other words, the online community is most effective if many participants generate and engage with posts.

2. Related Literature

2.1. Knowledge Sharing and Performance

This study is closely related to the literature on knowledge sharing and performance at the level of the organization and the individual. At the organization level, knowledge sharing refers to the process by which one unit (e.g., a team, regional subunit, or company) learns from the experiences of another unit (Argote and Ingram 2000, Dreyfus et al. 2020, Hwang and Krackhardt 2020). The relationship between organizational knowledge sharing (or learning) and organization-level performance has received significant attention from operations management scholars; most studies focus on how firms can improve performance by learning from other firms (e.g., Mani and Muthulingam 2019, Modi and Cantor 2021) or by facilitating knowledge sharing among subunits (Egelman et al. 2017). Baum and Dahlin (2007) examine how railroad firms can learn from their own accidents as well as other firms' accidents. In addition, the knowledge developed through one business's quality improvement initiatives can have spillover benefits for other businesses under some conditions (Muthulingam and Agrawal 2016). Other studies investigate how knowledge sharing among different subunits within the same firm can improve productivity. For instance, a manufacturing facility can improve productivity by promoting knowledge sharing and learning between shifts (Epple et al. 1996) and between different product generations (Egelman et al. 2017). Importantly, knowledge sharing does not necessarily happen uniformly throughout a company—for instance, in the context of pizza stores, Darr et al. (1995) show that knowledge is transferred across stores owned by the same franchise but not across stores owned by different franchisees, even though all stores belong to the same company.

At the individual level, knowledge sharing can affect individual performance (e.g., Cummings 2004, Qiu et al. 2017). In a public sector organization, knowledge-sharing behaviors mediate the relationship

between one's knowledge-sharing propensity and individual performance (Henttonen et al. 2016). Kang et al. (2008) indicate that individual work performance is positively related to the effectiveness of knowledge sharing. It is unclear, however, whether these demonstrated benefits translate to a geographically dispersed organization and whether the addition of a knowledge-sharing channel, such as an internal online community, affects individual job performance in such a setting. The increasing popularity of the multiunit organizational form makes it essential to understand whether the experience and knowledge gained by one subunit can be shared with and learned by peers at other subunits (Argote and Miron-Spektor 2011).

We propose that the introduction of a knowledge-sharing channel such as an online community can improve individual job performance in a geographically dispersed organization. Existing studies show that subunits within an organization can learn from the experiences of other subunits in the same organization (e.g., Darr et al. 1995, Epple et al. 1996). Sales representatives can use the online community to access knowledge that is not available locally but can be shared by peers from different geographic areas. An increase in the number and diversity of knowledge-sharing sources can improve outcomes. For example, projects involving experts from different locations tend to reap higher net earnings than projects that rely exclusively on local experts (Boh et al. 2007). Cummings (2004) shows that knowledge sharing between different groups (vs. knowledge sharing within the same group) has a stronger positive association with individual performance. In our context, sales representatives work in different geographic areas, but all representatives sell similar products to similar types of customers, so they should require similar knowledge to perform well. Learning about the products, selling experiences, and marketing techniques from peers in other subunits can help sales representatives improve their job performance.

2.2. Online Communities and Knowledge Management Systems

Many companies are now using public social media platforms, such as public online communities, to improve business operations (e.g., Allon and Babich 2020, Chan et al. 2016). Cui et al. (2018) show that using publicly available social media information can improve the accuracy of daily sales forecasts. Hong et al. (2018) show that activities on social media sites can drive an increase in fundraising for crowdfunding campaigns. Companies can learn from users' feedback in public online communities and adopt effective managerial response strategies to improve customer satisfaction (Gu and Ye 2014). Online communities also can be effective for individuals who are

pursuing health goals such as weight loss (Yan 2020) or chronic disease management (Liu et al. 2020). Our study contributes to this stream of literature by examining how internal knowledge management systems can improve individual job performance.

With the advancement of information technologies, companies are adopting internal knowledge management systems (e.g., a social media platform or online community) with the goal of capturing and sharing knowledge across the organization. A knowledge management system allows knowledge to be captured in a centralized manner that can be accessed through the Internet (e.g., Ko and Dennis 2011). Studies on the effects of *traditional* knowledge management systems have documented mixed findings regarding individual job performance (e.g., Handzic 2009, Ko and Dennis 2011), and scholars are just beginning to understand whether knowledge management systems with social elements can lead to beneficial outcomes. While traditional knowledge management systems restrict who (e.g., managers or IT department) can collect and distribute knowledge materials, systems with social elements, such as online communities, allow all participants to share content and react to existing content (e.g., Lu et al. 2015, Zhang 2017, Zhang and Venkatesh 2013).

While the existing research focuses on organizations with one centralized location, we study a geographically dispersed organization, in which the impacts of an online community are not yet understood. The effects within a centralized company may not necessarily translate to a geographically dispersed company if employees are more inclined to communicate with peers from the same location (vs. peers from other locations), leading to persistent knowledge silos (e.g., Hwang and Krackhardt 2020). In addition, knowledge and information that is useful in one geographic area may not be transferable to other locations. Thus, the net effect of an online community on individual performance in the geographically dispersed organization is an open question.

Our empirical analysis is also notable for its research design. Existing research focuses on the period after the knowledge management system has been introduced; researchers evaluate how the extent of community usage correlates with individual job performance. In this study, we leverage a quasi-natural experiment that spans one year *before* and one year *after* the introduction of the online community. Moreover, we use detailed records of community usage (instead of self-report surveys) and sales records (instead of supervisors' evaluations). Table 1 compares our study with existing studies on the effects of knowledge management systems.

We propose that an online community can improve individual job performance via two mechanisms: a knowledge sharing effect (described in earlier paragraphs) and a *social facilitation effect*. In social facilitation, an individual's performance is enhanced by the presence of other people (Falk and Ichino 2006, Zajonc 1965). It is widely documented that individuals on social media sites and online communities can be influenced by the presence of others. For instance, in an online learning community, students are more likely to exert effort and participate in tests if other students are doing the same task (Zhang et al. 2017). In the social-media-based prediction market, users' prediction accuracy is positively affected by the size of the social audience (Qiu and Kumar 2017). Huang et al. (2017) show that the integration of an online community with a social networking website can increase the production of user-generated content, indicating the importance of other people's presence for individual productivity. In our context, the social facilitation effect might occur if the online community heightens representatives' perception of the existence of colleagues in other geographic areas.

Similarly, individual performance is positively affected by peer effects and competition (e.g., Epple and Romano 1998) that tend to occur within physically collocated work teams (Chan et al. 2014). For

Table 1 Literature on Knowledge Management Systems and Individual Performance

Literature	Nature of the organization		Nature of the system		Research design	
	Geographically dispersed	Company employees	Knowledge sharing	Social elements	Methods	Performance measurement
McCall et al. (2008)					Lab experiment	Decision accuracy
Handzic (2009)					Lab experiment	Test scores
Quigley et al. (2007)			✓		Lab experiment	Market share in simulation
Ko and Dennis (2011)	✓	✓			Hierarchical linear modeling	Individual's sales
Zhang and Venkatesh (2013)		✓	✓	✓	Survey	Supervisor's rating
Lu et al. (2015)		✓	✓	✓	Structural equation	Supervisor's evaluation
Zhang (2017)		✓	✓	✓	Survey	Supervisor's rating
This Study	✓	✓	✓	✓	(1) Quasi-natural experiment (2) IV analysis	Individual's sales

example, Mas and Moretti (2009) show that the presence of high-ability grocery checkers increases the effort of nearby coworkers, and the effect involves social processes. In the context of an online community, online content generators become significantly more productive after the introduction of competing generators (Wang et al. 2019). We posit that the online community in our research context increases the effort (and performance) of sales representatives by enabling them to observe the job performance of colleagues from different geographic areas.

3. Research Context and Data

We obtain a unique data set from a large manufacturer in the sauce and condiment production industry in China to investigate how participation in a corporate internal online community affects the job performance of individual sales representatives. The company consists of one central factory for production and about a dozen geographically dispersed subunits for selling and marketing.² The subunits are distributed geographically across the country. The sales representatives in each subunit interact with and market to customers, most of whom are chefs of large restaurants, supermarket owners, or hotel managers. Each sales representative is assigned to one geographic territory and is responsible for all potential customers within his or her sales territory. In other words, sales representatives do not compete with their peers. Since career advancement opportunities and monetary bonuses are closely related to an individual's sales performance, the interests of the sales representatives are aligned well with the company's.

It is worth noting that the tasks of marketing and selling the company's products are knowledge-intensive. Each representative must interact with professionals and persuade them through testing and sampling events. For example, a representative may assist a restaurant chef with cooking two versions of a dish, one cooked with the company's product(s) and the other with a competitor's, to help the chef appreciate the difference. All new hires complete a series of trainings, and representatives are encouraged to share their selling experiences and techniques with their peers within the local subunit. At the beginning of 2018, the company introduced an online community to further facilitate communication and knowledge sharing among the representatives.

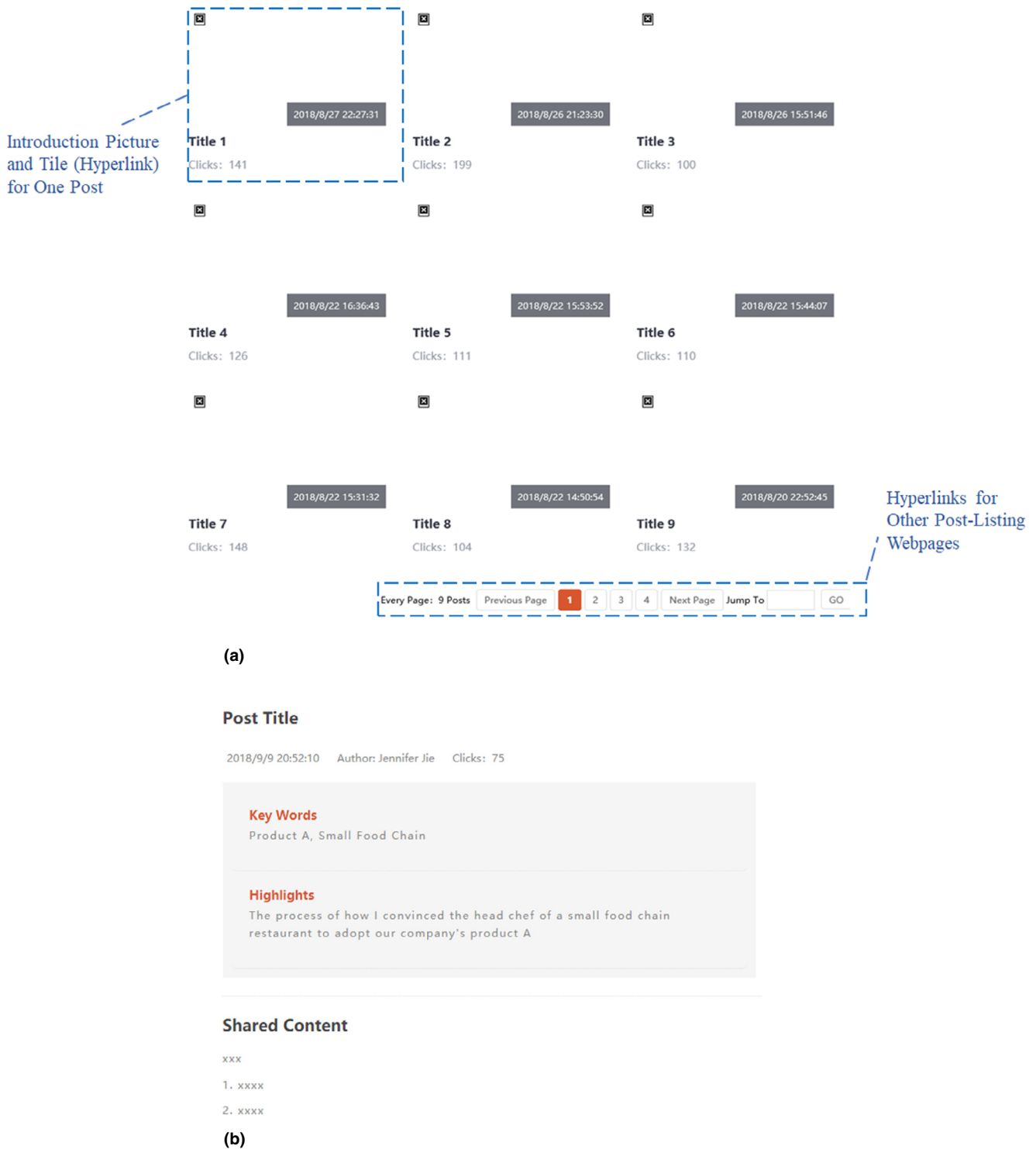
The corporate internal online community is a platform on which sales representatives can share their sales techniques, experiences, and stories based on one or more selling events. Representatives can generate new posts and comment on existing ones. The posts normally describe the company's product(s)

and related marketing/selling skills. All posts are accessible to all community participants. The company encourages representatives to use the online community, but using it is not mandatory. By the end of 2018 (i.e., the first year of the community's launch), an average of 42.6% of the representatives in each subunit participated in the community, and an average of 10.7% of the representatives in each subunit generated at least one post. All subunits had at least one representative who participated in the community, and about 85.0% of the subunits had a least one participant who generated a post. Most participants read posts generated by both local colleagues and nonlocal colleagues; among the posts that representatives read, 37.3% are generated by the colleagues from the local subunit.

The community comprises multiple post-listing webpages, and each webpage is laid out in a grid format with links to nine posts. Community users can access other post-listing webpages via hyperlinks. All posts are listed in inverse chronological order (i.e., the most recently generated posts appear on the first webpage). After representatives successfully log in to the community, the first page, which includes the most recent posts, loads. Representatives have access to all the posts; from the first page, representatives can navigate to additional post-listing webpages that contain older posts by clicking the page hyperlinks. On each post-listing webpage, representatives can click the title or introduction picture to view the post details on a separate post webpage. Figure 1, panel (a) shows a translated screenshot of the first post-listing webpage of the online community, taken on August 30, 2018.³ Figure 1, panel (b) illustrates a translated screenshot of the post webpage.

We work with three different departments within the company to build the data set. First, we collect the sales performance data from the marketing department. Because of the data's sensitivity, we are unable to access each sales record and customer information. Instead, we collect each representative's quarterly sales of all the company's products in the years 2017 and 2018 (Sales Data set 1) and monthly sales of one specific product (which we term "product-H") from August 2018 to December 2018 (Sales Data set 2). Second, we work with the company's IT department to collect detailed community usage records for each representative who participated in the online community during our study period. The records indicate when the representative logged in to the community, which post(s) he or she read during each visit to the community, and the posting time, topic information, and comments of the post(s). Third, we collaborate with the human resources department to collect each representative's demographic information, such as gender, age, subunit, and hire date.

Figure 1 Translated Screenshots of the Post-Listing Webpage and Post Webpage. Panel (a) One Post-Listing Webpage in the Online Community. Panel (b) One Post Webpage in the Online Community [Color figure can be viewed at wileyonlinelibrary.com]



We use Sales Data set 1 to conduct our main analysis with a quasi-natural experiment design; the data set includes each representative's sales performance both *before* and *after* the introduction of the online community at the beginning of 2018. Then, we

validate our main results with an instrumental-variable analysis on Sales Data set 2, which contains more granular sales records for one specific product (product-H) for 5 months *after* the introduction of the online community.

4. Main Analysis: A Quasi-Natural Experiment Design

We use a DID approach to test whether a corporate online community affects individual job performance in the geographically dispersed organization. Individual job performance is measured by individual quarterly sales of all the company's products (Sales Data set 1). In short, our DID analysis evaluates whether the sales performances of representatives who participated in the community changed before vs. after participation (i.e., treatment), with a group of nonparticipating representatives as the control for the unobservable time-variant factors. In the rest of this study, the “participating representatives” will be referred to as the “treated representatives” (logged in at least once during the study period), and the “nonparticipating representatives” will be referred to as the “control representatives” (never logged in during the study period). Figure 2 illustrates our DID design.

We use PSM to correct for a potential sample selection bias due to observable differences between the treated and control representatives. Specifically, each treated representative is paired with a control representative who is similar in terms of his or her probability of participating in the online community. We use one-to-one matching with replacement, under a caliper size 0.2 times the standard deviation of the propensity score (Xu et al. 2017). In one-to-one with replacement matching strategy, each treated representative is potentially matched with one control representative; each control representative may be matched with more than one treated representative (i.e., matching with replacement). The detailed matching procedures are in Appendix A. Table A.2 in Appendix A assesses the covariate balance after matching; the two matched groups have no statistical differences in most matching variables. Moreover, the distributions of propensity scores do not differ between the groups according to the Kolmogorov–Smirnov test ($p = 0.999$). Figure 3 shows that the

propensity score distributions of the treated and control representatives are similar after matching. We conclude that the treatment and control groups are adequately similar. In section 5.2.1, we verify the results by using alternative matching strategies.

Our matched samples include 240 representatives who participated in the online community (treatment group), along with the matched representatives who have not (control group). The validity of the DID framework relies on the parallel trends assumption—the treated and control representatives must have similar sales performance trends before the introduction of the online community. In section 5.2.2, we check this assumption with a leads-lags model and a placebo test; we find no significant difference between the treatment and control groups during the pre-treatment periods and a significant difference in the posttreatment periods.

We propose an econometric specification to estimate the impact of participation in the online community on the individual job performance of sales representatives. Specifically, we use the matched sample to build a panel data set at the representative-quarter (it) level. We use a fixed effects regression model to estimate the average treatment effect. Each observation conveys representative i 's product sales in quarter t . Our empirical model is

$$\begin{aligned} Sales_{it} = & \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 Treat_i \times Post_t \\ & + Control_{it} + \tau_t + \delta_i + \varepsilon_{it}, \end{aligned} \quad (1)$$

where $Sales_{it}$ is the dependent variable, representing the log-transformed sales made by representative i in quarter t .⁴ τ_t is a vector of calendar quarter dummies to control for unobservable time trends and seasonal effects, and δ_i captures representative-level fixed effects to control for unobservable time-invariant characteristics of the representative. $Treat_i$ is a binary variable that equals 1 if i joined the online community during our study period, and 0 otherwise. $Post_t$ is a binary variable that equals 0 if t

Figure 2 Difference-in-Differences Design [Color figure can be viewed at wileyonlinelibrary.com]

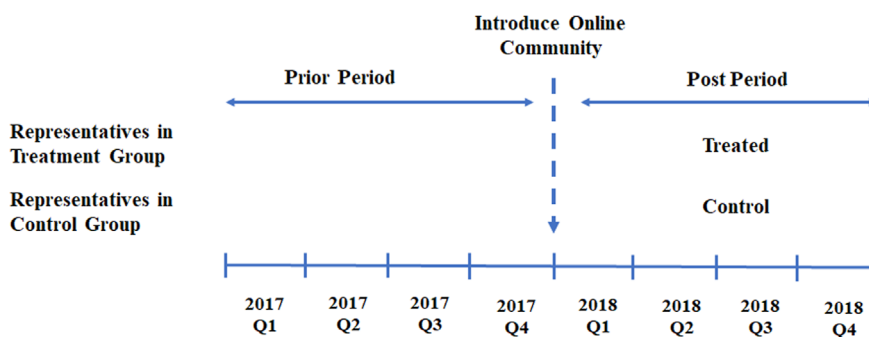
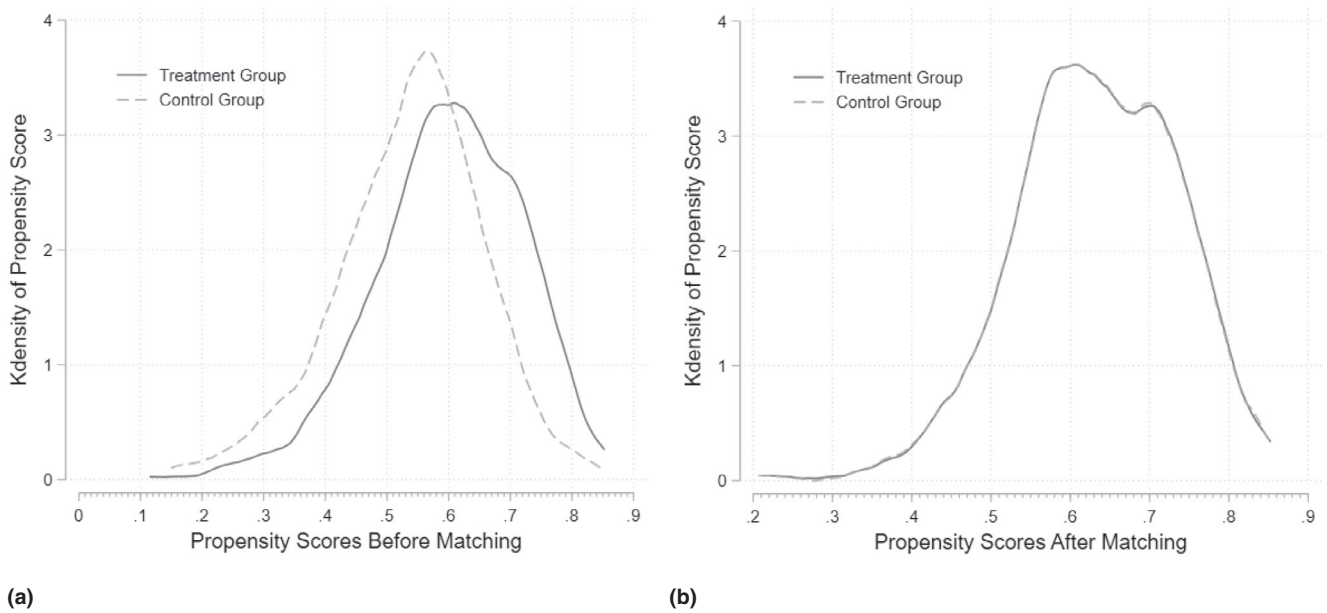


Figure 3 Distributions of Propensity Scores in the Treatment and Control Groups. Panel (a) Before Matching. Panel (b) After Matching



is in 2017 (before the online community was introduced) and 1 in 2018 (after its introduction). The coefficient of the interaction term, β_3 , is the DID estimator that captures how the individual sales performance changed after participation in the community. β_0 is the intercept, and ε_{it} is the mean-zero random error term.

$Control_{it}$ includes several control variables. We calculate representative i 's work experience, $WorkExp_{it}$, as the log-transformed number of months (before quarter t) since i joined the company. We control for the popularity of the company's products and brand with $Popularity_Pro_{it}$ and $Popularity_Brand_{it}$,⁵ the log-transformed Baidu search index of the company's products (summed for all products) and brand in i 's province in quarter t . We calculate popularity at the province-level because the index records suffer from high data sparsity (such that there are many zeroes at the city-level), and we believe the province-level captures the variance in popularity across the subunits' target markets. Lastly, we control for the population of representative i 's city at the end of quarter t 's year since sales are closely related to the population of the target market. We hand-collect the data from the Yearbooks of the National Bureau of Statistics of China or the Bureau of Statistics in i 's city.

Table 2 provides the description and statistics of the main variables in our DID analyses. Our panel data set is unbalanced because some representatives in our sample joined the company after the beginning of 2017, and some left the company before the end of 2018. As a robustness check, we re-estimate the model with a sample that includes only the representatives

who were active in the company during our whole study period. The results are consistent with those of our main analyses.

Table 3, Model 1 presents the model estimation without matching. The DID estimator is positive and significant ($\beta_3 = 0.222$; $p < 0.001$). The estimated coefficient has practical as well as statistical significance—it indicates that participation in the online community increases the representative's job performance by 24.86% (i.e., $e^{0.222} - 1$). Model 2 presents the model estimation with the matched sample, and the results are consistent with Model 1. In our context, the corporate online community provides an additional channel through which representatives can access new knowledge and observe the performance of colleagues from geographically dispersed subunits. The estimations indicate that in the geographically dispersed organization, allowing representatives from geographically dispersed subunits to observe and learn from each other in ways that improve their performance. We also examine the effect of community participation on representative i 's number of customers (log-transformed), and the estimation of β_3 is consistent with the values in Table 3.

5. Robustness Checks

In our main analysis, we use the company's introduction of the online community at the beginning of 2018 to design a quasi-natural experiment. We apply a DID model to the representatives' quarterly sales in 2017 and 2018, and we find an improvement in the sales performance of the representatives who participated

Table 2 Description of Variables in the Quasi-Natural Experiment

Variable	Description	Mean	SD	N
$Treat_i$	A binary variable indicating whether representative i participated in the online community: 1 = i participated 0 = i did not participate	0.496	0.500	2859
$Post_t$	A binary variable indicating whether the online community has been introduced: 1 = quarters after January 2018 0 = quarters before January 2018	0.517	0.5	2859
$Sales_{it}$	The log transformed sales that i makes in quarter t	6.886	1.419	2859
$WorkExp_{it}$	The log-transformed number of months i has worked in the company before quarter t	3.447	1.033	2859
$Popularity_Pro_{it}$	The log-transformed Baidu search index of the company's products at representative i 's province in quarter t	3.732	1.894	2859
$Popularity_Brand_{it}$	The log-transformed Baidu search index of the company's brand at representative i 's province in quarter t	4.609	1.981	2859
$CityPop_{it}$	The log-transformed total population (in ten thousand) at representative i 's city at the end of quarter t 's year	6.674	0.654	2859

in the online community (vs. representatives who did not participate). In this section, we conduct two groups of robustness checks. First, we use more detailed records of online community usage and a more granular level of sales records (i.e., Sales Data set 2) to design an instrumental-variable analysis. The results indicate that reading posts about one specific product in the community increases individual sales of that product. Second, we re-estimate our DID model with alternative matching strategies.

5.1. An Instrumental-Variable Analysis

Our main analysis uses the sales of all products at the representative-quarter level as the dependent variable. In this section, we use representative-level data on the monthly sales of one specific product (i.e., product-H) and detailed records of post content and community usage records to examine whether using the community affects individual job performance at a more granular level.

The community usage records allow us to observe each representative's community log-ins and the

Table 3 Impact of an Online Community on Individual Job Performance

	No matching (1) $Sales_{it}$	Propensity score matching (2) $Sales_{it}$
Model:		
Dep. var.:		
$Treat (\beta_1)$	—	—
$Post (\beta_2)$	0.144 (0.087)	−0.052 (0.107)
$Treat \times Post (\beta_3)$	0.222*** (0.076)	0.277*** (0.091)
$WorkExp$	−0.083 (0.081)	0.148* (0.087)
$Popularity_Pro$	0.034 (0.038)	0.070 (0.054)
$Popularity_Brand$	0.008 (0.014)	0.036* (0.022)
$CityPop$	8.362** (3.093)	11.473*** (3.337)
Quarter dummies	Yes	Yes
Representative FEs	Yes	Yes
R^2	0.659	0.522
Observations	5093	2859

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses. The estimation of the coefficient of $Treat$ is missing because of fixed effects at the representative level. The results are consistent if we do not include the quarter dummies or representative fixed effects. We do not include the result on the representative's number of customers for ease of exposition.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

posts they viewed during each visit, thereby enabling us to test whether reading the product-H-related posts affects the representative's sales of product-H. We focus on representatives who have sold product-H during the study period to make sure that selling product-H is within the scope of the representatives' work. We use the keyword tags provided by the post generator to check whether a specific post is relevant to product-H, and the company helps us confirm the accuracy of these tags. We construct a panel data set at the representative-month level in which each observation indicates representative i 's sales of product-H in month m and i 's post-reading behaviors before month m .

To test whether visiting product-H-related posts improves a representative's sales of product-H, it would be intuitive to create an econometric model in which representative i 's log-transformed sales of product-H in month m (i.e., $Sales_H_{im}$) is the dependent variable, and the log-transformed number of product-H-related posts he or she visited before month m (i.e., $PstRead_H_{im}$) is the independent variable. However, empirical identification would be challenging because of unobservable confounders. For example, i 's post-reading behaviors and sales performance may be simultaneously affected by the company's marketing focuses. Additionally, unobservable factors that are specific to each representative may also play a role in both online community usage and

job performance. We address time-invariant confounders with fixed effects at the representative level, and we address time-variant unobservable confounders with an exogenous instrumental variable for our independent variable of interest.

Specifically, we leverage the community's unique webpage design and use the log-transformed average number of product-H-related posts that i could have encountered before month m , $PstEncounter_Avg_H_{im}$, as the instrumental variable of $PstRead_H_{im}$. We construct this instrumental variable based on the fact that (i) every time representative i logs in to the online community, he or she sees the first post-listing webpage and (ii) the first post-listing webpage contains only the nine most recent posts, as described in section 3 and Figure 1, panel (a). To calculate $PstEncounter_Avg_H_{im}$, we first determine representative i 's total number of log-ins before month m , N_{im} . For each log-in l , we count the posts on the first post-listing webpage that are about product-H, $PstEncounter_H_{il}$. Then, $PstEncounter_Avg_H_{im}$ is the log-transformed average of $PstEncounter_H_{il}$ (the average of $PstEncounter_H_{il}$ is calculated as $\sum_{l=1}^{N_{im}} PstEncounter_H_{il} / N_{im}$). The logic of this instrumental variable is that posts with better positions—in this case, on the first post-listing page—are the most likely to be clicked (e.g., Guan and Cutrell 2007, Narayanan and Kalyanam 2015). Thus, when $PstEncounter_Avg_H_{im}$ is higher, representative i is more likely to encounter and read product-H-related posts when logging in to the online community, leading to a higher $PstRead_H_{im}$. However, $PstEncounter_Avg_H_{im}$ is unlikely to relate directly to i 's sales performance because most posts are generated by others, and whether a certain kind of post is on i 's first post-listing webpage depends on both the time at which the post generators publish the posts and the time at which i logs in to the community. Formally, we estimate a 2SLS regression:

$$PstRead_H_{im} = \alpha_0 + \alpha_1 PstEncounter_Avg_H_{im} + \alpha_2 WorkExp_{im} + \alpha_3 Popularity_H_{im} + \tau_{im}, \quad (2)$$

$$Sales_H_{im} = \gamma_0 + \gamma_1 PstRead_H_{im} + \gamma_2 WorkExp_{im} + \gamma_3 Popularity_H_{im} + \gamma_i + \varepsilon_{im}, \quad (3)$$

where Equation (2) reflects the first-stage regression, and Equation (3) reflects the second-stage regression. The dependent variable, $Sales_H_{im}$, is the log-transformed sales of product-H made by representative i in month m . The independent variable, $PstRead_H_{im}$, is the log-transformed number of product-H-related posts i read before month m . The control variables include $WorkExp_{im}$, the log-transformed number of months (before month m) since representative i joined the

company, and $Popularity_H_{im}$, the log-transformed Baidu search index of product-H in i 's province in month m .⁶ Table 4 provides the descriptions and statistics of the variables in this analysis.

We use a panel 2SLS estimation. As expected, the first-stage estimation in Table 5 shows a positive relationship between our instrumental variable $PstEncounter_Avg_H_{im}$ and independent variable $PstRead_H_{im}$, indicating that the instrument relevance criterion is met. We also examine the instrument strength by considering the Cragg–Donald Wald F statistic (363.004), which is well above the Stock–Yogo critical values for 2SLS relative biases of 5% (Stock and Yogo 2005). The Kleibergen and Paap (2006) rk Lagrange multiplier statistics yield a consistent result (186.66; $p < 0.001$). Model 1 in Table 6 presents the second-stage estimation results. The estimation of γ_1 is positive and statistically significant ($p < 0.001$), indicating that reading product-H-related posts increases representative i 's sales of product-H. We also estimate the model without using the instrumental variable (Table 6, Model 2), and the result is consistent.

We use $PstEncounter_Avg_H_{im}$ instead of the log-transformed cumulative sum of $PstEncounter_H_{il}$ as the instrumental variable because the number of i 's log-ins (i.e., N_{im}) can be endogenous. This instrumental variable is not perfect, however, because $PstEncounter_Avg_H_{im}$ is a real-time variable while $PstRead_H_{im}$ is a cumulative variable; the results have

Table 4 Descriptions of Variables in the Instrumental-Variable Analysis

Variable	Description	Mean	SD	N
$Sales_H_{im}$	The log-transformed sales of product-H that representative i made in month m	0.863	1.012	2139
$PstRead_H_{im}$	The log-transformed number of product-H-related posts read by representative i before month m	0.290	0.709	2139
$WorkExp_{im}$	The log-transformed number of months (before month m) since representative i joined the company	3.055	1.205	2139
$Popularity_H_{im}$	The log-transformed Baidu search index of product-H in representative i 's province in month m	5.305	0.358	2139
$PstEncounter_Avg_H_{im}$	The log-transformed average number of product-H-related posts i could have encountered during each log-in before month m	0.420	0.580	1861

Table 5 Instrument Relevance

Model:	(1)
Dep. var.:	<i>PstRead_H_{im}</i>
<i>PstEncounter_Avg_H</i>	0.450*** (0.028)
<i>WorkExp</i>	0.099*** (0.008)
<i>Popularity_H</i>	−0.337*** (0.024)
<i>R</i> ²	0.154
Observations	1861

*** $p < 0.01$.

different interpretations. In addition, we must focus on representatives who logged in to the online community at least once (i.e., $N_{im} > 0$) or else the average calculation would be undefined. To overcome the shortcomings and show the robustness of this analysis, we construct two additional instrumental variables. First, we use the log-transformed cumulative sum of *PstEncounter_H_{it}* as the instrumental variable. Second, we construct a cumulative instrumental variable that uses the number of product-H-related posts encountered by *i* in one randomly-selected visit in each month before *m*. The results with these two instrumental variables are consistent with the results in Table 6, Model 1. Appendix B describes the detailed procedures and results.

5.2. Robustness Checks with Alternative Experiment Designs

We perform a series of robustness checks by varying the experiment designs with alternative matching approaches and by examining the parallel trends assumption with a leads-lags model and a placebo test. Our main result remains consistent across these robustness checks.

Table 6 Product sales and Reading Posts About Product-H

Model:	FE/2SLS (1)	FE (2)
Dep. var.:	<i>Sales_H_{im}</i>	<i>Sales_H_{im}</i>
<i>PstRead_H</i> (γ_1)	0.402*** (0.112)	0.110** (0.050)
<i>WorkExp</i>	0.190 (0.173)	0.419*** (0.095)
<i>Popularity_H</i>	−0.016 (0.262)	−0.356 (0.232)
Representative FEs	Yes	Yes
Observations	1853	2139

Note: The unit of analysis is at the representative-month level. Robust standard errors are in parentheses. There are more observations in Model 2 than in Model 1 because *PstEncounter_Avg_H* (instrumental variable) is missing when a representative has zero log-ins. Eight singleton observations are dropped in Model 1.

** $p < 0.05$, *** $p < 0.01$.

5.2.1. Alternative Matching Approaches. The main model uses one-to-one PSM with replacement. In this section, we test three alternative matching approaches (Pu et al. 2017): one-to-one PSM *without* replacement, PSM with the nearest-three-neighbors approach, and coarsened exact matching (CEM). In PSM *without* replacement, a control representative is not available for matching after he or she is matched with a treated representative. In the nearest-three-neighbors matching approach, one treated representative can be matched with up to three control representatives. We further use CEM to construct our matched samples. While PSM matches treated and control representatives based on their probabilities of joining the online community, as determined by the pretreatment variables, CEM creates bins or ranges for each matching variable, and observations with sufficiently close values are assigned to the same bins and marked as matched (Iacus et al. 2012, Kumar et al. 2019). After CEM, the data set contains 262 representatives, of whom 147 participated in the online community. The multivariate imbalance tests show that CEM improves the balance level $\mathcal{L}_1$ of the data set from 0.830 to 0.688.

We re-estimate Equation (1) with the three newly constructed matched samples, and the results are reported in Table 7. (Note that the strategies result in different sample sizes.) The findings in all three models are consistent with those in Table 3—that is, the coefficient of the DID estimator is positive and significant, validating our finding that participation in the online community improves individual sales performance.

5.2.2. Tests to Examine the Parallel Trends Assumption. The company in our context introduced the online community at the beginning of 2018, and we construct the panel data set for our main analysis with representative-level data spanning four quarters before and four quarters after the treatment. It is possible, however, that the treated representatives gradually increased their sales performance for unknown reasons even before the introduction of the online community, while the control representatives did not. In this case, the treatment effect identified in our DID estimation would be due to a pre-existing divergence in the sales trend rather than to the online community. We adopt a leads-lags model to show that the sales performance of the treated representatives does not differ from the performance of the control representatives in the pretreatment periods. Following Jung et al. (2019) and Dhanorkar and Muthulingam (2020), we verify the absence of an unusual pre-existing trend with a two-way, fixed-effects model that includes a group of dummy variables for the time periods before and after the introduction of the corporate online community:

$$Sales_{it} = \alpha_0 + \sum_{j=t-4}^{t+3} \alpha_j(s_i\varphi) + Control_{it} + \tau_t + \delta_i + \varepsilon_{it}, \quad (4)$$

where $Sales_{it}$ is the dependent variable, indicating representative i 's log-transformed sales in quarter t ; τ_t is a vector of calendar quarter dummies; δ_i captures representative fixed effects; s_i is a binary variable indicating whether representative i participated in the online community in our study period; and φ is a vector of time dummies representing the chronological distance between the observation period and the beginning of 2018 (i.e., when the online community was launched). Additionally, to avoid a dummy variable trap, we choose the quarter before the launch time as the base quarter and omit the dummy variable for it (Dhanorkar and Muthulingam 2020, Jung et al. 2019).

Table 8 presents the results of the leads-lags model. All estimations with time dummies in the pre-treatment periods are insignificant (i.e., for $t - 4$, $t - 3$, and $t - 2$), indicating pretreatment homogeneity between the treated and control representatives. In the posttreatment periods, however, the treated representatives have a higher average sales performance than the control representatives, consistent with the estimations of our main analyses. In Appendix C, following Kumar et al. (2019), we further examine the parallel trends assumption via a placebo test with

reconstructed study time windows. The results help validate our conclusions with a shorter time window and rule out the possibility of a pre-existing divergence in the trends of the two groups of representatives.

6. Additional Analyses

Based on the literature on knowledge sharing and online communities, we propose that a corporate online community improves individual job performance in the geographically dispersed organization for two main reasons. First, the online community enables representatives to access the experience and knowledge shared by colleagues from other geographically dispersed subunits. Second, the online community provides an environment where users can better sense the existence of their peers and observe the effort and experiences of peers who perform similar tasks, thereby leading to a social facilitation or peer effect. In this section, we conduct additional analyses to provide evidence for these mechanisms.

6.1. Heterogeneous Effects on Sales Representatives

After establishing the main effects of the online community on individual job performance, we examine some interesting moderating factors and explore characteristics that enable some representatives to benefit more than others from the introduction of the online community. An online community in the geographically dispersed organization provides an additional

Table 7 Alternative Matching Approaches

Model: Dep. var.:	One-to-one propensity score matching (PSM) (without replacement) (1) $Sales_{it}$	Nearest three neighbors PSM (with replacement) (2) $Sales_{it}$	Coarsened exact matching (3) $Sales_{it}$
<i>Treat</i> (β_1)	—	—	—
<i>Post</i> (β_2)	−0.131 (0.115)	0.042 (0.072)	−0.068 (0.147)
<i>Treat</i> × <i>Post</i> (β_3)	0.325*** (0.104)	0.215*** (0.067)	0.370*** (0.128)
<i>WorkExp</i>	0.217** (0.101)	0.145** (0.065)	−0.037 (0.145)
<i>Popularity_Pro</i>	0.058 (0.068)	0.126*** (0.048)	0.071 (0.071)
<i>Popularity_Brand</i>	0.046* (0.104)	0.062*** (0.016)	−0.044** (0.021)
<i>CityPop</i>	16.127** (3.843)	14.701*** (2.676)	9.040** (3.935)
Quarter dummies	Yes	Yes	Yes
Representative FEs	Yes	Yes	Yes
R^2	0.517	0.526	0.399
Observations	2200	5573	1599

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses. The estimated coefficient of *Treat* is missing because of the fixed effects at the representative level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8 Leads-Lags Model for Testing the Parallel Trends Assumption

Model: Dep. var.:	(1) $Sales_{it}$
$t - 4$	−0.113 (0.125)
$t - 3$	−0.208 (0.158)
$t - 2$	0.007 (0.116)
$t - 1$	Base quarter
$t + 0$	0.316*** (0.120)
$t + 1$	0.268** (0.135)
$t + 2$	0.258** (0.131)
$t + 3$	0.253*** (0.093)
Control variables	Yes
Quarter dummies	Yes
Representative FEs	Yes
R^2	0.522
Observations	2859

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses.

** $p < 0.05$, *** $p < 0.01$.

channel for representatives to observe and learn from the shared experiences of peers in different geographic areas. Therefore, it is possible that the online community should be *more* beneficial for representatives in subunits with *fewer* offline learning opportunities. To test this theory, we generate two variables: $LocSubSize_{it}$, the log-transformed number of peer representatives at representative i 's subunit, reflects the extent of i 's opportunities to converse with and learn from colleagues offline. $LocAvgPreSales_{it}$, the log-transformed average sales performance of colleagues at representative i 's subunit before the introduction of the online community, can proxy the quality of local colleagues' offline communication.

We examine the heterogeneous effects by adding three-way interaction terms to Equation (1). As predicted, the estimations of the three-way interaction terms are significant and negative, indicating that the online community is more beneficial for representatives in subunits with fewer employees (Table 9, Model 1) and for those in subunits with lower average job performance (Model 2). The results verify our conjecture that the online community offers an additional channel through which representatives can observe and learn from peers dispersed across different locations.⁷

6.2. The Effects of Different Types of Posts

In our main analysis, we built a classic DID model with two binary variables to represent the time period (before vs. after the introduction of the online community) and the group (control vs. treatment); we estimated the average treatment effect of the online community on individual sales performance. In this subsection, we further examine the relationship between a representative's post reading behaviors and sales performance by creating a new independent variable: the log-transformed number of posts that representative i read before quarter t (i.e., $PstRead_Total_{it}$). The new independent variable captures variation in community usage both across representatives and over time. Different from the analyses in section 6.1, here, we cannot examine the relationship between $PstRead_Total_{it}$ and $Sales_{it}$ in the DID framework by adding a three-way interaction term. $PstRead_Total_{it}$ is highly correlated with $Treat_i \times Post_t$ (i.e., the DID estimator) because $PstRead_Total_{it}$ is positive only if $Treat_i \times Post_t$ equals 1 (correlation coefficient: 0.797). Thus, following the existing literature (e.g., Acemoglu et al. 2004), we use the following model:

$$Sales_{it} = \beta_0 + \beta_1 PstRead_Total_{it} + \beta_2 Post_t + \beta_3 Control_{it} + \tau_t + \delta_i + \varepsilon_{it}, \quad (5)$$

where $Post_t$ and $Control_{it}$ are as defined in section 4. Consistent with our main result, the estimation results (Table 10, Model 1) indicate that i 's sales in

Table 9 Moderation by Opportunities at the Local Subunit

Model: Dep. var.:	(1) <i>Sales_{it}</i>	(2) <i>Sales_{it}</i>
<i>Treat</i>	—	—
<i>Post</i>	−0.133 (0.114)	−0.069 (0.108)
<i>Treat × Post</i>	0.335*** (0.095)	0.276*** (0.091)
<i>LocSubSize</i>	—	—
<i>Post × LocSubSize</i>	0.398*** (0.093)	—
<i>Treat × Post × LocSubSize</i>	−0.415*** (0.107)	—
<i>LocAvgPreSales</i>	—	—
<i>Post × LocAvgPreSales</i>	—	0.259*** (0.059)
<i>Treat × Post × LocAvgPreSales</i>	—	−0.347*** (0.090)
<i>WorkExp</i>	0.189** (0.089)	0.173* (0.089)
<i>Popularity_Pro</i>	0.032 (0.058)	0.065 (0.054)
<i>Popularity_Brand</i>	0.035 (0.021)	0.025 (0.021)
<i>CityPop</i>	8.818*** (3.337)	10.138*** (3.311)
Quarter dummies	Yes	Yes
Representative FEs	Yes	Yes
R^2	0.527	0.525
Observations	2859	2859

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses. Estimations for the coefficients of *Treat*, *LocSubSize*, and *LocAvgPreSales* are missing because of the fixed effects at the representative level. *LocSubSize* and *LocAvgPreSales* are standardized.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

quarter t are positively related to the number of posts he or she read in the community before quarter t .⁸

As discussed in our theorization, one unique characteristic of the online community in a geographically dispersed organization is that participants are from different subunits. A natural question is whether representatives are differentially affected by reading posts generated in the local vs. nonlocal subunits. On the one hand, if knowledge and experience are not transferrable across geographic boundaries, then the online community may primarily encourage communication among individuals from the same subunit (e.g., Hwang et al. 2015). On the other hand, if knowledge and experience are transferrable, then the online community may enable representatives to access information that is unavailable locally (Tallman and Phene 2007). Knowledge sharing between different groups (vs. within the same group) can be more strongly associated with individual performance (Cummings 2004). It is also possible that the online presence of representatives from other geographic subunits exerts a greater peer effect (i.e., is more effective at motivating representatives to exert more

effort), given that the performance of local colleagues is already observable.

We formally examine these possibilities with a model that differentiates each post read by representative i based on whether the author of the post is from i 's subunit. Specifically, we generate two independent variables, $PstRead_Local_{it}$ and $PstRead_NonLocal_{it}$: the log-transformed number of posts read by i before quarter t that were generated by colleagues from i 's subunit and by colleagues from all other subunits, respectively. We alter Equation (5) by substituting $PstRead_Total_{it}$ with one of these two independent variables. The results are presented in Table 10, Models 2 and 3.⁹ The estimations indicate that i 's performance has a significantly positive association with the number of nonlocal posts that i read (Model 3) but only a marginally positive association with the number of local posts that i read (Model 2). The results imply that the online community in the geographically dispersed organization has the potential to improve individual performance because it creates opportunities for knowledge sharing and social facilitation across geographic areas. Even though users of the online community are still disproportionately inclined to communicate with their local peers, and knowledge silos may still exist in online communities (e.g., Hwang and Krackhardt 2020), we show that an online community helps individuals improve their job performance by providing an additional channel

where they can communicate with nonlocal colleagues. We also examine the moderating role of work experience on the effect of reading nonlocal posts by adding the interaction term $PstRead_NonLocal_{it} \times WorkExp_{it}$. The estimation of the interaction term is positive and significant (Table 10, Model 4), indicating that reading nonlocal posts can be more beneficial for representatives who have more work experience.

We theorized that participation in the online community affects individual sales performance via two mechanisms: the knowledge sharing effect and the social facilitation effect. We test this theory by differentiating between posts that describe a single selling event (i.e., single-narrative post) and posts that describe multiple selling events (i.e., multisummary post). In a single-narrative post (illustrated in Figure 4, panel a), the author provides background information and describes conversations with the customer. In a multisummary post (illustrated in Figure 4, panel b), the author explains how they used one selling strategy to succeed in multiple selling events, and they disclose their sales performance as evidence of the strategy's efficacy. Both the knowledge sharing and social facilitation effects would predict a stronger effect of multisummary posts (vs. single-narrative posts) on individual job performance because the multisummary post contains knowledge that is better summarized and organized, and the inclusion of the author's sales performance can better stimulate peer effects and

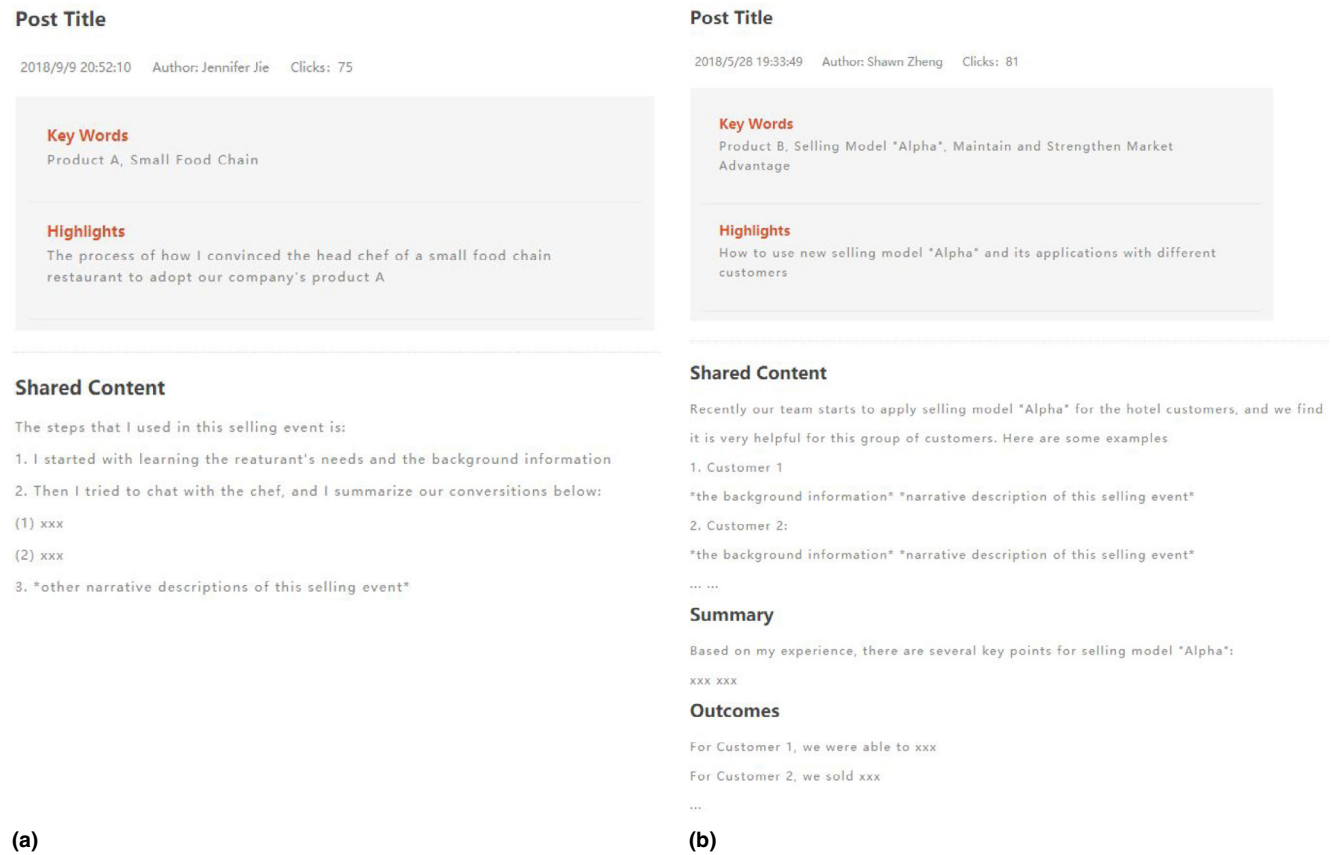
Table 10 The Effects of Local vs. Nonlocal Posts

Model: Dep. var.:	(1) <i>Sales_{it}</i>	(2) <i>Sales_{it}</i>	(3) <i>Sales_{it}</i>	(4) <i>Sales_{it}</i>
<i>PstRead_Total</i>	0.063** (0.029)	—	—	—
<i>PstRead_Local</i>	—	0.004 (0.046)	—	—
<i>PstRead_NonLocal</i>	—	—	0.079** (0.031)	−0.190 (0.115)
<i>PstRead_NonLocal</i> × <i>WorkExp</i>	—	—	—	0.076** (0.032)
<i>Post</i>	0.022 (0.101)	0.084 (0.096)	0.015 (0.101)	−0.017 (0.105)
<i>WorkExp</i>	0.151* (0.087)	0.156* (0.086)	0.156* (0.087)	0.195** (0.095)
<i>Popularity_Pro</i>	0.067 (0.053)	0.066 (0.054)	0.066 (0.053)	0.064 (0.054)
<i>Popularity_Brand</i>	0.038* (0.022)	0.042* (0.022)	0.042* (0.022)	0.039* (0.022)
<i>CityPop</i>	12.088*** (3.36)	12.353*** (3.333)	12.938*** (3.438)	12.939*** (3.433)
Quarter dummies	Yes	Yes	Yes	Yes
Representative FEs	Yes	Yes	Yes	Yes
R^2	0.521	0.520	0.521	0.521
Observations	2859	2859	2859	2859

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 Translated Screenshots of Two Types of Posts. Panel (a) A Single-Narrative Post. Panel (b) A Multisummary Post [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/poms.13565)]



competition, potentially motivating representatives to exert more effort (Wang et al. 2019).

We test this theory by generating two additional independent variables, $PstRead_SigNarrative_{it}$ and $PstRead_MulSummary_{it}$, which are the log-transformed numbers of single-narrative and multisummary posts read by i before quarter t . We replace $PstRead_Total_{it}$ in Equation (5) with each new independent variable, separately.¹⁰ The results in Table 11 indicate that i 's performance is significantly and positively associated with the number of multi-summary posts that i read (Model 2) but not significantly associated with the number of single-narrative posts that i read (Model 1). The results confirm our conjectures and provide some empirical support for our theorization. We then examine the moderating role of work experience on the effect of reading multi-summary posts by adding the interaction term $PstRead_MulSummary_{it} \times WorkExp_{it}$. The estimation of the interaction term is insignificant (Table 11, Model 3), indicating that the relationship between reading multisummary posts and job performance is not moderated by work experience.

The purpose of this study is to investigate the effects of an online community on job performance in

the geographically dispersed organization. We acknowledge two limitations of our additional analyses that explore the underlying mechanisms. First, these analyses only serve to check whether the empirical evidence is aligned with our proposed mechanisms; we cannot eliminate the unobservable factors (e.g., work effort) that possibility affect both online community engagement and individual sales performance. Second, our analyses do not disentangle the knowledge sharing and social facilitation effects because, in our context, there is no post type that contains information about only sales performance (i.e., for social facilitation) or only selling event(s) (i.e., for knowledge sharing). In addition, within posts that contain both types of information, we cannot discern which part of each post receives more of the representative's attention. We infer that both effects are likely to occur simultaneously when representatives visit posts, and we cannot discern which part of the post receives the most of representatives' attention. Nevertheless, to the best of our knowledge, we are among the first to investigate the effects of the corporate internal online community on individual job performance in the geographically dispersed organization.

Table 11 The Effects of Single-narrative vs. Multisummary Posts

Model: Dep. var.:	(1) <i>Sales_{it}</i>	(2) <i>Sales_{it}</i>	(3) <i>Sales_{it}</i>
<i>PstRead_SigNarrative</i>	0.014 (0.034)	—	—
<i>PstRead_MulSummary</i>	—	0.060** (0.029)	−0.054 (0.101)
<i>PstRead_MulSummary</i> × <i>WorkExp</i>	—	—	0.033 (0.028)
<i>Post</i>	0.078 (0.096)	0.039 (0.098)	0.026 (0.102)
<i>WorkExp</i>	0.155* (0.086)	0.150* (0.087)	0.172* (0.096)
<i>Popularity_Pro</i>	0.065 (0.054)	0.068 (0.053)	0.067 (0.054)
<i>Popularity_Brand</i>	0.041* (0.022)	0.038* (0.022)	0.038* (0.022)
<i>CityPop</i>	12.406*** (3.400)	12.019*** (3.361)	11.786*** (3.382)
Quarter dummies	Yes	Yes	Yes
Representative FEs	Yes	Yes	Yes
<i>R</i> ²	0.520	0.520	0.521
Observations	2859	2859	2859

Note: The unit of analysis is at the representative-quarter level. Robust standard errors are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We are also among the first to explore the differential effects of reading posts from different subunits and reading different types of posts on the community participants.

7. Conclusion and Discussion

Corporate online communities are increasingly popular in the geographically dispersed organization, but there has been little empirical evidence of the benefits for individual job performance in such a setting. In this study, we use a quasi-natural experiment to quantify the effect of an internal online community on individual job performance in the geographically dispersed organization. The unique data set from a large manufacturer enables us (i) to examine the effect of an online community using observations both *before* and *after* its introduction, (ii) to design an instrumental-variable analysis using detailed community usage records, (iii) to explore the heterogeneous nature of the effect across individual sales representatives, and (iv) to examine the effects of different types of posts. Econometric analyses and robustness checks confirm that in the geographically dispersed organization, participating in a corporate online community and reading posts from the community positively affect individual job performance. In addition, we find that the online community is more beneficial for representatives who have fewer offline opportunities to observe and learn from local colleagues (operationalized as the number

of employees and average job performance at the representative's subunit). Finally, we show that individual job performance is more positively related to reading posts generated by nonlocal colleagues (vs. posts generated by local colleagues) and posts that summarize multiple selling events (vs. posts that describe one selling event).

This study makes several contributions to the prior operations management and information systems literature. First, this study contributes to the literature on knowledge sharing and performance (e.g., Darr et al. 1995, Mani and Muthulingam 2019). While the extant studies show that knowledge sharing and learning can improve performance at the organization and individual levels, little is understood about whether the addition of a knowledge-sharing channel, such as an online community, can improve individual job performance in the geographically dispersed organization. Our work suggests that when representatives participate in the online community, the access to the shared knowledge and work experiences of peers from geographically dispersed areas enables the representatives to perform their jobs more effectively.

Second, our work extends prior studies on online communities. Most existing studies focus on (i) public social media, such as public online communities, where users are customers (Gu and Ye 2014, Hu et al. 2019) or healthcare service users (Liu et al. 2020, Yan 2020), or (ii) knowledge-sharing patterns in online communities (e.g., Hwang et al. 2015, Pu et al. 2020, 2021). In this study, we are interested in the effects of a corporate internal online community on the performance of participants in that community. Our results imply that an internal online community should be considered an effective tool for connecting employees who otherwise are separated geographically. The community heightens participants' awareness of peers, leading to an improved job performance through the social facilitation effect.

Third, this study adds to the literature on the relationship between individual job performance and knowledge management systems (e.g., Ko and Dennis 2011). Unlike traditional knowledge management systems, corporate online communities incorporate social media elements and motivate knowledge sharing among all participants. While the existing studies focus on organizations with one centralized location, we study the effects of an online community in a geographically dispersed organization. Our results indicate that participation in the online community confers heterogeneous benefits that are influenced by characteristics of the local subunit, such as its size and average sales performance.

Our study is subject to several limitations that present future research opportunities. First, our narrow

focus on a manufacturer within one industry (sauces and condiments) means that our results have limited generalizability. In our context, knowledge is likely to be transferrable and applicable across dispersed locations because all representatives sell the same types of products and interface with similar types of customers. Therefore, the conclusions of this study are more likely to apply to geographically dispersed organizations in which all subunits do similar types of work. It would be interesting for future researchers to consider the effects of a corporate online community in a different context. Second, we operationalize job performance as sales performance because the company in our context focuses on sales. However, different companies may have different goals, so individual job performance may need to be quantified in another way in a different context. Third, as discussed in section 6.2, the types of posts in our setting do not enable us to disentangle the knowledge sharing and social facilitation effects because the two effects are likely to occur simultaneously when representatives read multisummary posts. Future studies can use a different setting or conduct surveys to examine these two effects separately.

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Notes

¹“Knowledge management system” refers to “a class of information systems applied to managing organizational knowledge” (Alavi and Leidner 2001, p. 114). The system can facilitate knowledge sharing and transfer among employees within the organization, and a knowledge management system represents a type of social technology (Zhang 2017). The internal online community is one emerging type of knowledge management system.

²We are not able to provide potentially identifiable information of the company such as the exact number of subunits.

³All the original texts in this online community are in Chinese. To protect the company’s identity, we edited the screenshots by removing or obfuscating the product name, post pictures, and detailed content.

⁴In this paper, “log-transformed” means that we add 1 to the variable and then take the natural logarithm. We use the log-transformed values of the variables because these variables are positive and count in nature.

⁵Baidu is the largest search engine in China, and the search index of a single word can represent the number of individuals who searched that word during a specified time period. We collect the search index for all the company’s products (*Popularity_Pro_{it}*) and brand (*Popularity_Brand_{it}*) at the province-quarter level. See the detailed

description of this index at <http://www.baiduguide.com/baidu-seo-guide/baidu-keyword-trends/> (accessed March 22, 2021).

⁶Note that *Popularity_H_{im}* refers to the Baidu search index of product-H at the province-month level, while the popularity control variable in the main analysis refers to the summed Baidu search indices of all products at the province-quarter level.

⁷We also use *WorkExp_{it}* as a moderating factor to examine whether the effect of the online community is heterogeneous across the representatives’ work experience. The estimation shows that the effect of the online community is not moderated by *WorkExp_{it}*. We do not include the detailed estimations here for ease of exposition.

⁸We also add the quadratic term of *PstRead_Total_{it}* to examine the nonlinear relationship between sales performance and post reading behaviors. The estimation of the quadratic term is significant and negative, indicating that the effect of reading posts diminishes as representatives read more and more posts. We do not include the detailed estimations here for ease of exposition.

⁹We do not include *PstRead_Local_{it}* and *PstRead_NonLocal_{it}* in the same regression test because they are highly correlated (correlation coefficient: 0.483), suggesting a potential harmful effect of collinearity (e.g., Mela and Kopalle 2002).

¹⁰*PstRead_SigNarrative_{it}* is highly correlated with *PstRead_MulSummary_{it}* (correlation coefficient: 0.689), so we do not include them in the same regression to avoid the potential multicollinearity issue.

References

- Acemoglu, D., D. H. Autor, D. Lyle. 2004. Women, war, and wages: The effect of female labor supply on the wage structure at midcentury. *J. Polit. Econ.* **112**(3): 497–551.
- Alavi, M., D. E. Leidner. 2001. Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Q.* **25**(1): 107–136.
- Allon, G., V. Babich. 2020. Crowdsourcing and crowdfunding in the manufacturing and services sectors. *Manuf. Serv. Oper. Manag.* **22**(1): 102–112.
- Almeida, P., B. Kogut. 1999. Localization of knowledge and the mobility of engineers in regional networks. *Management Sci.* **45**(7): 905–917.
- Argote, L., P. Ingram. 2000. Knowledge transfer: A basis for competitive advantage in firms. *Organ. Behav. Hum. Decis. Process.* **82**(1): 150–169.
- Argote, L., E. Miron-Spektor. 2011. Organizational learning: From experience to knowledge. *Organ. Sci.* **22**(5): 1123–1137.
- Baum, J., K. Dahlin. 2007. Aspiration performance and railroads’ patterns of learning from train wrecks and crashes. *Organ. Sci.* **18**(3): 368–385.
- Boh, W. F., Y. Ren, S. Kiesler, R. Bussjaeger. 2007. Expertise and collaboration in the geographically dispersed organization. *Organ. Sci.* **18**(4): 595–612.
- Bughin, J., M. Chui. (2013). Evolution of the networked enterprise. *McKinsey Digital* (March). Available at <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/evolution-of-the-networked-enterprise-mckinsey-global-survey-results> (accessed date August 27, 2021).
- Bughin, J., M. Chui, H. Martin (2016). How social tools can reshape the organization. *McKinsey Digital* (May). Available at <https://www.mckinsey.com/business-functions/mckinsey->

- digital/our-insights/how-social-tools-can-reshape-the-organization (accessed date August 27, 2021).
- Chan, H. K., X. Wang, E. Lacka, M. Zhang. 2016. A mixed-method approach to extracting the value of social media data. *Prod. Oper. Manag.* **25**(3): 568–583.
- Chan, T. Y., J. Li, L. Pierce. 2014. Compensation and peer effects in competing sales teams. *Management Sci.* **60**(8): 1965–1984.
- Cui, R., S. Gallino, A. Moreno, D. J. Zhang. 2018. The operational value of social media information. *Prod. Oper. Manag.* **27**(10): 1749–1769.
- Cummings, J. N. 2004. Work groups, structural diversity, and knowledge sharing in a global organization. *Management Sci.* **50**(3): 352–364.
- Darr, E. D., L. Argote, D. Eppler. 1995. The acquisition, transfer, and depreciation of knowledge in service organizations: Productivity in franchises. *Management Sci.* **41**(11): 1750–1762.
- Dhanorkar, S., S. Muthulingam. 2020. Do e-waste laws create behavioral spillovers? Quasi-experimental evidence from California. *Prod. Oper. Manag.* **29**(7): 1738–1766.
- Dreyfus, D., A. Nair, S. Talluri. 2020. The impact of chain organization size on efficiency and quality of affiliated facilities – Implications for multi-unit organizational forms. *Prod. Oper. Manag.* **29**(7): 1605–1623.
- Egelman, C. D., D. Eppler, L. Argote, E. R. Fuchs. 2017. Learning by doing in multiproduct manufacturing: Variety, customizations, and overlapping product generations. *Manage. Sci.* **63**(2): 405–423.
- Eppler, D., L. Argote, K. Murphy. 1996. An empirical investigation of the microstructure of knowledge acquisition and transfer through learning by doing. *Oper. Res.* **44**(1): 77–86.
- Eppler, D., R. E. Romano. 1998. Competition between private and public schools, vouchers, and peer-group effects. *Am. Econ. Rev.* **88**(1): 33–62.
- Falk, A., A. Ichino. 2006. Clean evidence on peer effects. *J. Labor Econ.* **24**(1): 39–57.
- Gu, B., Q. Ye. 2014. First step in social media: Measuring the influence of online management responses on customer satisfaction. *Prod. Oper. Manag.* **23**(4): 570–582.
- Guan, Z., E. Cutrell. 2007. An eye tracking study of the effect of target rank on web search. B. Begole, S. Payne, E. Churchill, R. Sr. Amant, D. Gilmore, M. D. Rosson, eds. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, 417–420.
- Hagel, J., J. S. Brown. (2008). Harrah's new twist on prediction markets. *Bloomberg*, December 22. Available at <https://www.bloomberg.com/news/articles/2008-12-22/harrahs-new-twist-on-prediction-marketsbusinessweek-business-news-stock-market-and-financial-advice> (accessed date August 27, 2021).
- Handzic, M. 2009. Evaluating KMS effectiveness for decision support: Preliminary results. W. R. King, ed. *Knowledge Management and Organizational Learning*. Springer US, Boston, MA, 236–250.
- Henttonen, K., A. Kianto, P. Ritala. 2016. Knowledge sharing and individual work performance: An empirical study of a public sector organization. *J. Knowl. Manag.* **20**(4): 749–768.
- Hong, Y., Y. Hu, G. Burtch. 2018. Embeddedness, prosociality, and social influence: Evidence from online crowdfunding. *MIS Q.* **42**(4): 1211–1224.
- Hu, Y., A. Xu, Y. Hong, D. Gal, V. Sinha, R. Akkiraju. 2019. Generating business intelligence through social media analytics: Measuring brand personality with consumer-, employee-, and firm-generated content. *J. Manage. Inf. Syst.* **36**(3): 893–930.
- Huang, N., Y. K. Hong, G. Burtch. 2017. Social network integration and user content generation: Evidence from natural experiments. *MIS Q.* **41**(4): 1035–1058.
- Hwang, E. H., D. Krackhardt. 2020. Online knowledge communities: Breaking or sustaining knowledge silos? *Prod. Oper. Manag.* **29**(1): 138–155.
- Hwang, E. H., P. V. Singh, L. Argote. 2015. Knowledge sharing in online communities: Learning to cross geographic and hierarchical boundaries. *Organ. Sci.* **26**(6): 1593–1611.
- Iacus, S., G. King, G. Porro. 2012. Causal inference without balance checking: coarsened exact matching. *Polit. Anal.* **20**(1): 1–24.
- Jung, J., R. Bapna, J. Ramaprasad, A. Umyarov. 2019. Love unshackled: Identifying the effect of mobile app adoption in online dating. *MIS Q.* **43**(1): 57–72.
- Kang, Y. J., S. E. Kim, G. W. Chang. 2008. The impact of knowledge sharing on work performance: An empirical analysis of the public employees' perceptions in South Korea. *Int. J. Public Admin.* **31**(14): 1548–1568.
- Kleibergen, F. R., R. Paap. 2006. Generalized reduced rank tests using the singular value decomposition. *J. Econ.* **133**(1): 97–126.
- Ko, D.-G., A. R. Dennis. 2011. Profiting from knowledge management: The impact of time and experience. *Inf. Syst. Res.* **22**(1): 134–152.
- Kumar, A., A. Mehra, S. Kumar. 2019. Why do stores drive online sales? Evidence of underlying mechanisms from a multichannel retailer. *Inf. Syst. Res.* **30**(1): 319–338.
- Liu, Q. B., X. Liu, X. Guo. 2020. The effects of participating in a physician-driven online health community in managing chronic disease: Evidence from two natural experiments. *MIS Q.* **44**(1): 391–419.
- Lu, B., X. Guo, N. Luo, G. Chen. 2015. Corporate blogging and job performance: Effects of work-related and nonwork-related participation. *J. Manage. Inf. Syst.* **32**(4): 285–314.
- Mani, V., S. Muthulingam. 2019. Does learning from inspections affect environmental performance? Evidence from unconventional well development in Pennsylvania. *Manuf. Serv. Oper. Manag.* **21**(1): 177–197.
- Mas, A., E. Moretti. 2009. Peers at work. *Am. Econ. Rev.* **99**(1): 112–145.
- McCall, H., V. Arnold, S. G. Sutton. 2008. Use of knowledge management systems and the impact on the acquisition of explicit knowledge. *J. Inf. Syst.* **22**(2): 77–101.
- Mela, C. F., P. K. Kopalle. 2002. The impact of collinearity on regression analysis: The asymmetric effect of negative and positive correlations. *Appl. Econ.* **34**(6): 667–677.
- Modi, S. B., D. E. Cantor. 2021. How coopetition influences environmental performance: Role of financial slack, leverage, and leanness. *Prod. Oper. Manag.* **30**(7): 2046–2068.
- Muthulingam, S., A. Agrawal. 2016. Does quality knowledge spillover at shared suppliers? An empirical investigation. *Manuf. Serv. Oper. Manag.* **18**(4): 525–544.
- Narayanan, S., K. Kalyanam. 2015. Position effects in search advertising and their moderators: A regression discontinuity approach. *Market. Sci.* **34**(3): 388–407.
- Pu, J., Y. Chen, L. Qiu, H. K. Cheng. 2020. Does identity disclosure help or hurt user content generation? Social presence, inhibition, and displacement effects. *Inf. Syst. Res.* **31**(2): 297–322.
- Pu, J., Y. Kwark, S. Han, B. Gu, Q. Ye. 2017. The double-edged sword of expert reviewer programs: The effects of offering expert reviewer status on review generation. Y. J. Kim, R. Agarwal, J. K. Lee, eds. *Proceedings of the International Conference on Information Systems (ICIS 2017)*. Association for Information Systems, Atlanta, 1–17. Digital Platforms, Paper 10.
- Pu, J., Y. Liu, Y. Chen, L. Qiu, H. K. Cheng. 2021. What questions are you inclined to answer? Effects of hierarchy in corporate knowledge sharing communities. *Inf. Syst. Res.* Forthcoming.

- Qiu, L., H. K. Cheng, J. Pu. 2017. Hidden profiles in corporate prediction markets: The impact of public information precision and social interactions. *MIS Q.* **41**(4): 1249–1273.
- Qiu, L., S. Kumar. 2017. Understanding voluntary knowledge provision and content contribution through a social-media-based prediction market: A field experiment. *Inf. Syst. Res.* **28**(3): 529–546.
- Quigley, N. R., P. E. Tesluk, E. A. Locke, K. M. Bartol. 2007. A multilevel investigation of the motivational mechanisms underlying knowledge sharing and performance. *Organ. Sci.* **18**(1): 71–88.
- Stock, J. H., M. Yogo. 2005. Testing for weak instruments in linear IV regression. D. W. K. Andrews, J. H. Stock, eds. *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*. Cambridge University Press, New York, 80–108.
- Tallman, S., A. Phene. 2007. Leveraging knowledge across geographic boundaries. *Organ. Sci.* **18**(2): 252–260.
- Wang, Y., P. Goes, Z. Wei, D. Zeng. 2019. Production of online word-of-mouth: Peer effects and the moderation of user characteristics. *Prod. Oper. Manag.* **28**(7): 1621–1640.
- Xu, K., J. Chan, A. Ghose, S. P. Han. 2017. Battle of the channels: The impact of tablets on digital commerce. *Management Sci.* **63**(5): 1469–1492.
- Yan, L. 2020. The kindness of commenters: An empirical study of the effectiveness of perceived and received support for weight-loss outcomes. *Prod. Oper. Manag.* **29**(6): 1448–1466.
- Zajonc, R. B. 1965. Social facilitation. *Science* **149**(3681): 269–274.
- Zhang, D. J., G. Allon, J. A. Van Mieghem. 2017. Does social interaction improve learning outcomes? Evidence from field experiments on massive open online courses. *Manuf. Serv. Oper. Manag.* **19**(3): 347–367.
- Zhang, X. 2017. Knowledge management system use and job performance: A multilevel contingency model. *MIS Q.* **41**(3): 811–840.
- Zhang, X., V. Venkatesh. 2013. Explaining employee job performance: The role of online and offline workplace communication networks. *MIS Q.* **37**(3): 695–722.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix A: Steps of propensity score matching.

Appendix B: Alternative instrumental variables.

Appendix C: A placebo test.